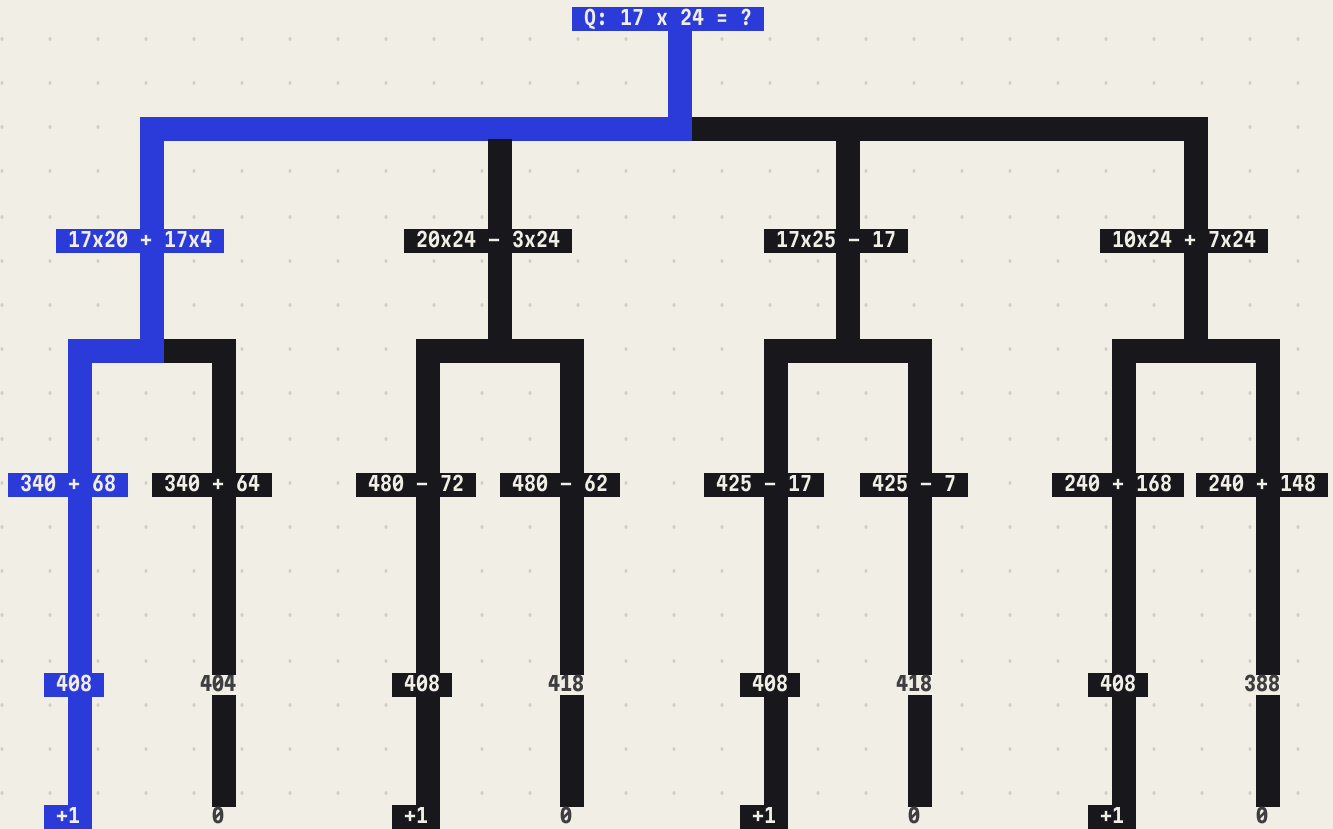




```
reward = 1 if answer == 408 else 0    •    advantage = reward - mean(group)
```

```
loss = -(advantage * logprob(chain)).mean()    # sample a group, reinforce the right ones
```

Charlie and the Reasoning Room

Reinforcement Learning, from bandits to reasoning models